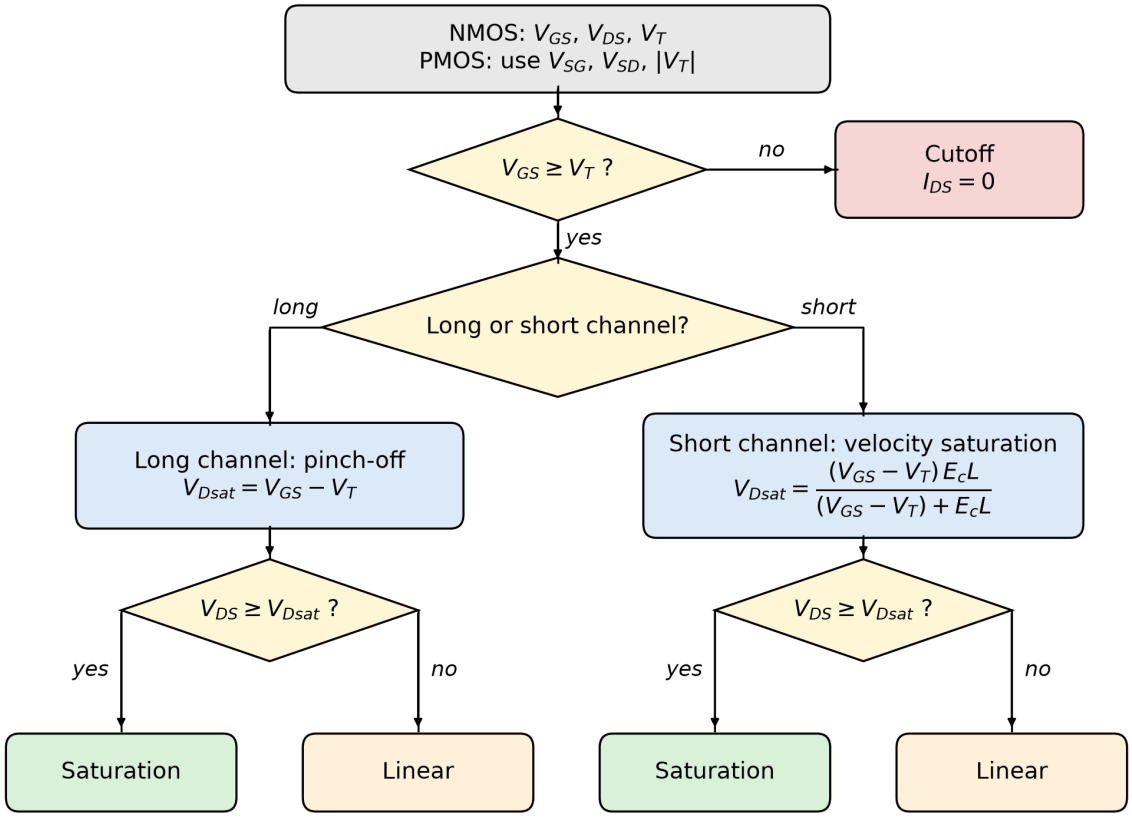
Exercise 04 – Solution Summary

Task 4.1

Idea: Find the region of operation from the terminal voltages: first V_{GS} vs. V_T, then V_{DS} vs. $V_{GS}-V_T$.

Given: NMOS, $V_T = 0.4\,\text{V}$.

Flowchart: region of operation



Current equations (P81, summary):

Channel	Region	I_{DS}
both	Cutoff	0
Long	Linear	$\frac{k}{2}\left[2(V_{GS}-V_T)V_{DS} - V_{DS}^2\right]$
Long	Saturation	$\frac{k}{2}(V_{GS}-V_T)^2(1+\lambda V_{DS})$
Short	Linear	$\frac{W}{L}\,\frac{\mu_e C_{ox}}{1+V_{DS}/(E_cL)}\left(V_{GS}-V_T-\frac{V_{DS}}{2}\right)V_{DS}$
Short	Saturation	$W v_{sat} C_{ox}\,\frac{(V_{GS}-V_T)^2}{(V_{GS}-V_T)+E_cL}$

- $k = \mu C_{ox}\dfrac{W}{L}$, $\quad v_{sat} = \dfrac{\mu E_c}{2}$, $\quad \mu_e$: effective mobility (Task 3.1 (e))
- Short channel with $E_cL \to \infty \to$ long-channel equations

Solution (a)–(d):

- NMOS: S bottom, D up.

	V_{GS}	$V_{GS} \geq V_T$?	V_{DS}	$V_{GS} - V_T$	$V_{DS} \geq V_{GS} - V_T$?	Region
a)	$0.2\,\text{V}$	no	–	–	–	**cutoff**
b)	$0.1\,\text{V}$	no	–	–	–	**cutoff**
c)	$1.2\,\text{V}$	yes	$0.2\,\text{V}$	$0.8\,\text{V}$	no	**linear**
d)	$0.8\,\text{V}$	yes	$1.0\,\text{V}$	$0.4\,\text{V}$	yes	**saturation**

Homework: replace all NMOS with PMOS and determine the region

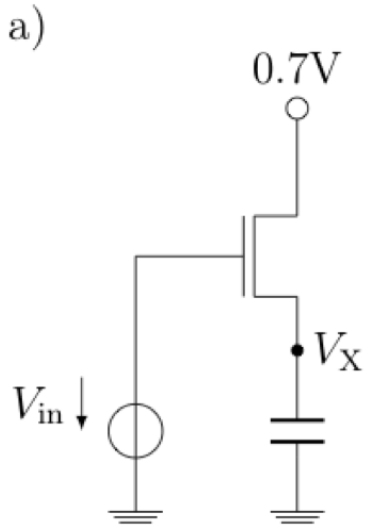
hint: $V_{Tp} = -0.4\,\text{V}$, S up, D down, use V_{SG}, V_{SD}.

Task 4.2

Idea: The capacitor charges through the NMOS until the current stops: either V_{GS} drops to V_T (cutoff) or V_{DS} drops to 0. Steady state: V_X where $I_{DS} = 0$.

Given: $V_T = 0.4\,\text{V}$, $0 < V_{in} < 1.2\,\text{V}$, capacitor initially discharged, body effect neglected.

1. **Task (a): gate = V_{in}, drain = 0.7 V, source = V_X (source follower)**



   - **Source follower** (official: voltage follower): input at the gate, output at the source → V_X follows V_{in}, shifted down by V_T; used as buffer / level shifter
   - $V_{in} \leq 0.4\,\text{V}$: $V_{GS} = V_{in} < V_T$ → cutoff → $V_X = 0$
   - $0.4\,\text{V} < V_{in} \leq 1.1\,\text{V}$: NMOS on, charges C until $V_{GS} = V_{in} - V_X = V_T$:

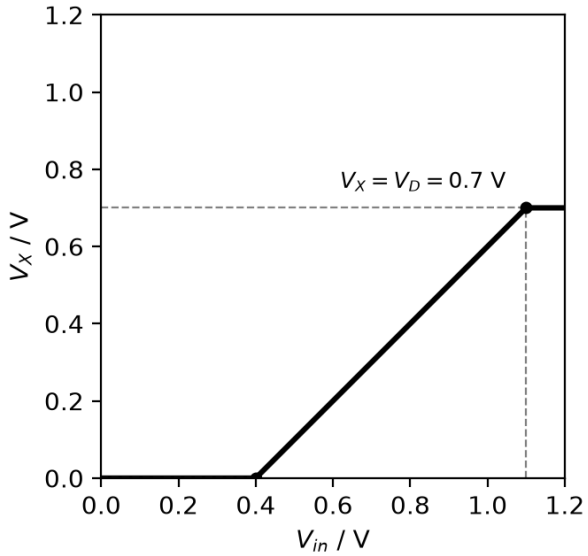
$$V_X = V_{in} - V_T$$

- $V_{in} > 1.1\,\text{V}$: V_X can't exceed the drain voltage → linear, no charging anymore:

$$V_X = V_D = 0.7\,\text{V}$$

- Together:

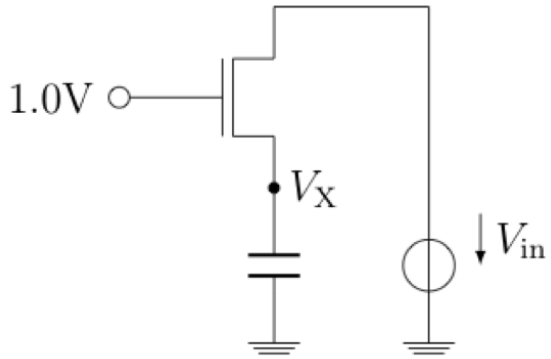
$$\boxed{V_X = \min\left(\max(V_{in} - V_T,\, 0),\; 0.7\,\text{V}\right)}$$



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2. **Task (b): gate = $1.0\,\text{V}$, drain = V_{in}, source = V_X (pass transistor)**

b)



- **Pass transistor** (official: current follower): gate = control signal, the input is passed from drain to source when the NMOS is on → switch; used in pass-transistor logic, transmission gates, SRAM access transistors
- $V_{in} \leq 0.6\,\text{V}$: $V_{GS} = 1.0\,\text{V} - V_X \geq V_T$ → on, charges until $V_{DS} = 0$:

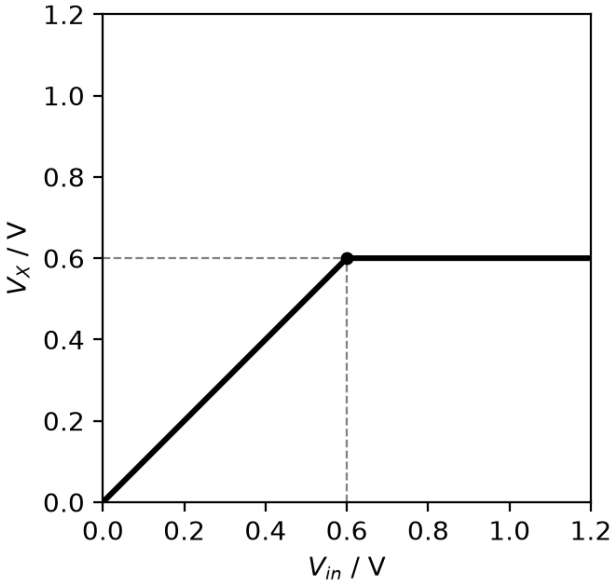
$$V_X = V_{in}$$

- $V_{in} > 0.6\,\text{V}$: $V_{GS} = 1.0\,\text{V} - V_X = V_T$, no more charging:

$$V_X = V_G - V_T = 1.0\,\text{V} - 0.4\,\text{V} = 0.6\,\text{V}$$

- Together:

$$\boxed{V_X = \min\left(V_{in},\ V_G - V_T\right)}$$



Note:

- NMOS passes a “strong 0” but only a “weak 1”: output limited to $V_G - V_T$ (threshold drop)
- With body effect ($V_{SB} = V_X > 0$) V_T increases $\rightarrow$ V_X even lower

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Task 4.3

Idea: Intrinsic gate capacitance $C_G = WLC_{ox}$, split into C_{GS}, C_{GD}, C_{GB} depending on the region.

Given: PMOS, $t_{ox} = 22\,\text{Å}$, $W = 400\,\text{nm}$, $L = 100\,\text{nm}$, cutoff at $V_{GS} = 0\,\text{V}$.

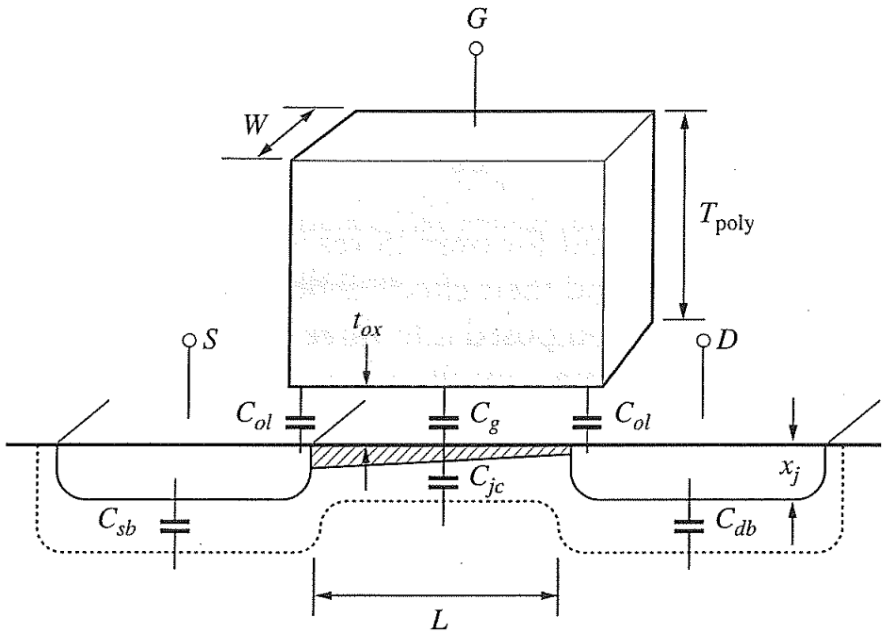


Figure 2.17
Capacitances of the MOS transistor.

1. **Gate capacitance per width**, equation (2.34), P71:

$$C_g = L\,C_{ox} = L\,\frac{\epsilon_{ox}}{t_{ox}}, \qquad \epsilon_{ox} = 4\,\epsilon_0$$

$$C_g = 100 \cdot 10^{-7}\,\text{cm} \cdot \frac{4 \cdot 8.85 \cdot 10^{-14}\,\text{AsV}^{-1}\text{cm}^{-1}}{22 \cdot 10^{-8}\,\text{cm}} \approx 1.6 \cdot 10^{-11}\,\text{F/cm} = 1.6\,\text{fF}/\mu\text{m}$$

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2. **Total gate capacitance**

$$C_G = W\,C_g = 0.4\,\mu\text{m} \cdot 1.6\,\text{fF}/\mu\text{m} = \boxed{0.64\,\text{fF}}$$

- $C_G = C_{GS} + C_{GD} + C_{GB}$

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3. **Split by region**, Fig. 2.18, P72:

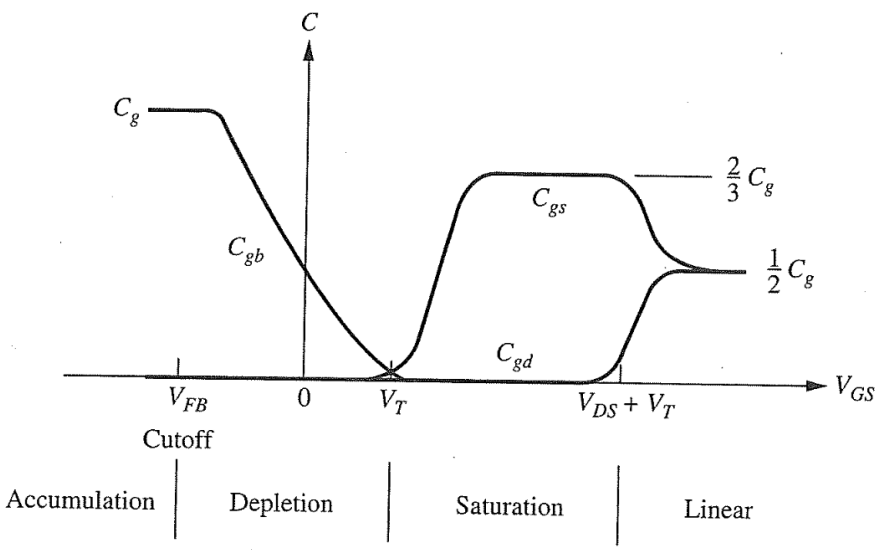


Figure 2.18
Capacitances based on region of operation.

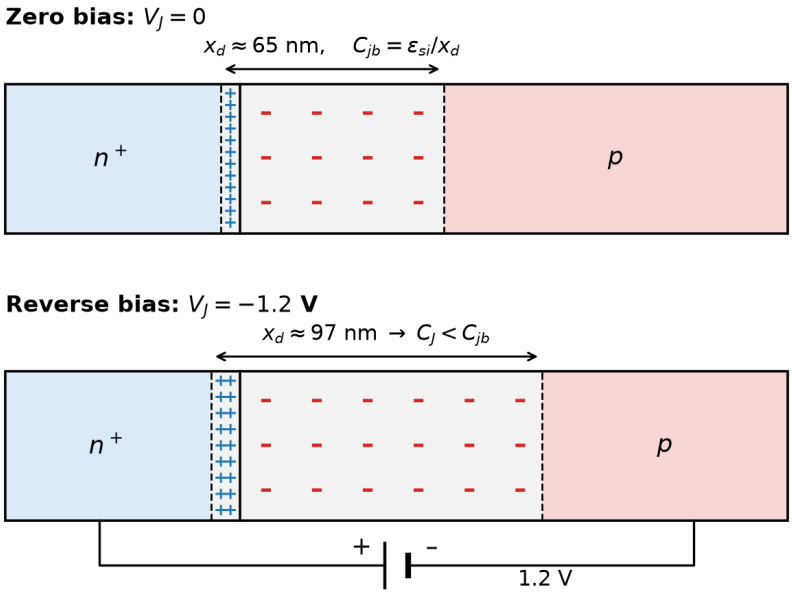
Region	C_{GS}	C_{GD}	C_{GB}	Reason
Cutoff ($V_{GS} = 0$)	0	0	$\frac{1}{2}C_G = 0.32\,\text{fF}$	no channel → gate sees bulk
Linear	$\frac{1}{2}C_G = 0.32\,\text{fF}$	$\frac{1}{2}C_G = 0.32\,\text{fF}$	0	channel shared by S and D
Saturation	$\frac{2}{3}C_G = 0.43\,\text{fF}$	0	0	pinch-off at D → only S

Task 4.4

Idea: Source/drain–bulk pn junction is a voltage-dependent capacitor: the wider the depletion region (reverse bias), the smaller C_J.

Given: n^+p junction, $N_D = 10^{20}\,\text{cm}^{-3}$, $N_A = 3 \cdot 10^{17}\,\text{cm}^{-3}$, $W = 400\,\text{nm}$, $L = 100\,\text{nm}$, $x_j = 50\,\text{nm}$, $Y = 300\,\text{nm}$.

1. **Task (a): Built-in potential ϕ_B and zero-bias capacitance C_{jb}**
   - ϕ_B: voltage across the depletion region of the junction at zero bias, set by the doping on both sides
   - C_{jb}: junction capacitance per unit area at zero bias ($V_J = V_p - V_n, V_J = 0$)



- Equation (2.37), P74:

$$\phi_B = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = 0.026\,\text{V} \cdot \ln \frac{3 \cdot 10^{17}\,\text{cm}^{-3} \cdot 10^{20}\,\text{cm}^{-3}}{(1.45 \cdot 10^{10}\,\text{cm}^{-3})^2} \approx \boxed{1\,\text{V}}$$

- Equation (2.39), P75:

$$C_{jb} = \sqrt{\frac{\epsilon_{si}\, q}{2\phi_B} \cdot \frac{N_A N_D}{N_A + N_D}}$$

- $N_D \gg N_A \rightarrow \frac{N_A N_D}{N_A + N_D} \approx N_A$ (one-sided junction), equation (2.40), P75:

$$C_{jb} \approx \sqrt{\frac{\epsilon_{si}\, q\, N_A}{2\phi_B}} = \sqrt{\frac{11.7 \cdot 8.85 \cdot 10^{-14}\,\text{AsV}^{-1}\text{cm}^{-1} \cdot 1.6 \cdot 10^{-19}\,\text{As} \cdot 3 \cdot 10^{17}\,\text{cm}^{-3}}{2 \cdot 1\,\text{V}}}$$

$$C_{jb} \approx 1.6 \cdot 10^{-7}\,\text{F/cm}^2 = \boxed{1.6\,\text{fF}/\mu\text{m}^2}$$

Note: $1\,\text{F/cm}^2 = 10^7\,\text{fF}/\mu\text{m}^2$

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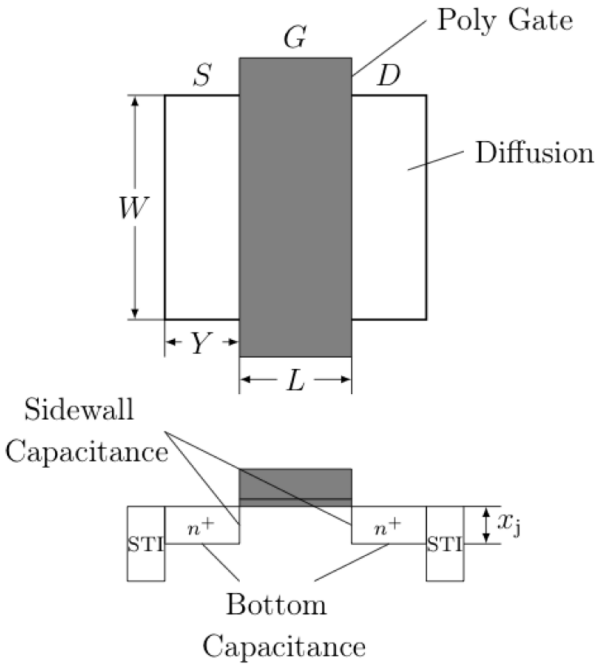
2. **Task (b):** C_J **for** $V_J = 0\,\text{V}$ **and** $V_J = -1.2\,\text{V}$

- Equation (2.38), P74:

$$C_J = \frac{C_{jb}\, A}{\left(1 - \frac{V_J}{\phi_B}\right)^m}$$

  - A: junction area, $m = 1/2$: grading coefficient (abrupt junction)
  - $V_J < 0$: reverse bias (e.g. drain at $1.2\,\text{V}$, bulk at $0\,\text{V}$)

- Junction area: bottom $Y \cdot W$ + sidewall towards the channel $x_j \cdot W$ (other sidewalls face STI):



$$A = (Y + x_j)\, W = (0.3\,\mu\text{m} + 0.05\,\mu\text{m}) \cdot 0.4\,\mu\text{m} = 0.14\,\mu\text{m}^2$$

- $V_J = 0\,\text{V}$:

$$C_J(0\,\text{V}) = C_{jb}\, A = 1.6\,\text{fF}/\mu\text{m}^2 \cdot 0.14\,\mu\text{m}^2 \approx \boxed{0.22\,\text{fF}}$$

- $V_J = -1.2\,\text{V}$:

$$C_J(-1.2\,\text{V}) = \frac{0.22\,\text{fF}}{\left(1 - \dfrac{-1.2\,\text{V}}{1\,\text{V}}\right)^{1/2}} = \frac{0.22\,\text{fF}}{\sqrt{2.2}} \approx \boxed{0.15\,\text{fF}}$$

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3. **Task (c): K_{eq} for $V_1 = -1.2\,\text{V}$, $V_2 = 0\,\text{V}$ (large signal)**
   - C_J changes during switching → use the average capacitance over the voltage swing: $C_{eq} = K_{eq}\, C_{jb}$
   - Equation (2.43), P77 ($m = 1/2$):

$$K_{eq} = \frac{-2\phi_B^{1/2}}{V_2 - V_1}\left[(\phi_B - V_2)^{1/2} - (\phi_B - V_1)^{1/2}\right]$$

$$K_{eq} = \frac{-2 \cdot (1\,\text{V})^{1/2}}{0\,\text{V} - (-1.2\,\text{V})}\left[(1\,\text{V})^{1/2} - (2.2\,\text{V})^{1/2}\right] \approx \boxed{0.8}$$

   - Equation (2.44), P77:

$$C_J = K_{eq}\, C_{jb}\, (Y + x_j)\, W = 0.8 \cdot 0.22\,\text{fF} \approx \boxed{0.18\,\text{fF}}$$

Note: $0.15\,\text{fF} < 0.18\,\text{fF} < 0.22\,\text{fF}$: large-signal C_J lies between the two end points

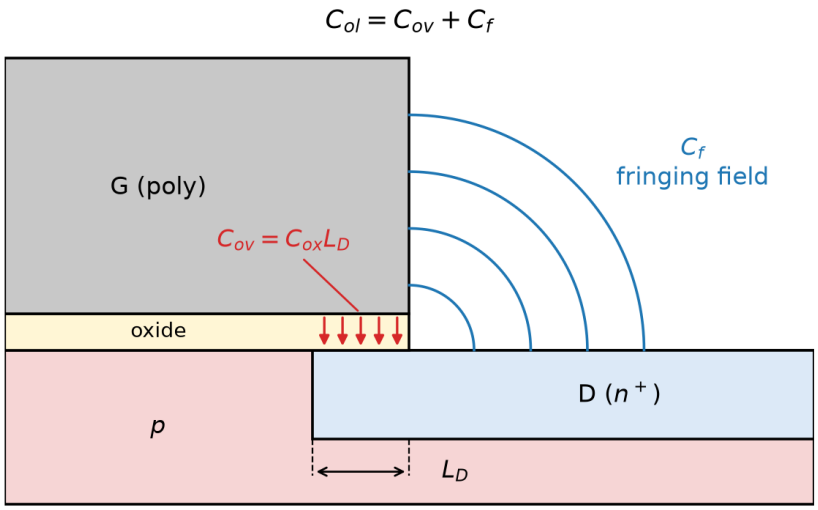
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Task 4.5

Idea: Gate overlaps S/D by the lateral diffusion L_D (parallel plate) plus fringing field from the poly sidewall: $C_{ol} = C_{ov} + C_f$.

Given: $0.13\,\mu\text{m}$ ($t_{ox} = 22\,\text{Å}$, $C_{ox} = 1.6\,\mu\text{F/cm}^2$), $T_{poly}/t_{ox} = 100$, $L_D = 10\,\text{nm}$.

Gate edge (not to scale):



1. **Fringing capacitance**, equation (2.47), P79:

$$C_f = \frac{2\epsilon_{ox}}{\pi} \ln\left(1 + \frac{T_{poly}}{t_{ox}}\right) = \frac{2 \cdot 4 \cdot 8.85 \cdot 10^{-14}\,\text{AsV}^{-1}\text{cm}^{-1}}{\pi} \ln(101) \approx 0.1\,\text{fF}/\mu\text{m}$$

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2. **Overlap (parallel plate)**, P80:

$$C_{ov} = C_{ox}\,L_D = 1.6 \cdot 10^{-6}\,\text{F/cm}^2 \cdot 10 \cdot 10^{-7}\,\text{cm} = 1.6 \cdot 10^{-12}\,\text{F/cm} = 0.16\,\text{fF}/\mu\text{m}$$

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3. **Overlap capacitance**, equation (2.46), P79:

$$C_{ol} = C_{ov} + C_f = 0.16\,\text{fF}/\mu\text{m} + 0.1\,\text{fF}/\mu\text{m} = \boxed{0.26\,\text{fF}/\mu\text{m}}$$